

THERMAL ENGINEERING-I*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): 1x20
 - a. The value of $n = 1$ in the polytropic process indicates it to be – (a) adiabatic process (b) isobaric process (c) isothermal process (d) isentropic process
 - b. Measurement of temperature is based on _____ law of thermodynamics – (a) zeroth (b) first (c) second (d) third
 - c. An open system is one in which _____. (a) Mass does not cross boundaries of the system, through energy may do so (b) Neither mass nor energy crosses the boundaries of the system (c) Both energy and mass cross the boundaries of the system (d) Mass crosses the boundary but not the energy
 - d. The first law of Thermodynamics relates to – (a) Conservation of mass (b) Conservation of momentum (c) Conservation of internal energy (d) Conservation of energy
 - e. Properties of substances like pressure, temperature and density, in thermodynamic coordinates are – (a) path functions (b) point functions (c) cyclic functions (d) real functions
 - f. “No process is possible whose sole result is the transfer of heat from a colder object to a hotter object” this statement is related to the – (a) zeroth law of thermodynamics (b) first law of thermodynamics (c) Kelvin Planck statement (d) Clausius statement
 - g. A machine which can supply mechanical work continuously without consumption of any energy is called as – (a) PMM1 (b) PMM2 (c) PMM3 (d) all the above
 - h. Boyle’s law relates to – (a) Constant volume (b) Constant pressure (c) Constant temperature (d) none of the above
 - i. For a perfect gas the specific heat at constant pressure C_p and that at constant volume C_v are related as – (a) $C_p = C_v$ (b) $C_p = C_v + R$ (c) $C_p = C_v - R$ (d) $C_p = C_v + \log R$
 - j. The statement that molecular weights of all gases occupy the same volume is known as – (a) Dalton’s law (b) Joule’s law (c) Boyle’s law (d) Avogadro’s hypothesis
 - k. The conditions at which a real gas behaves as an ideal gas – (a) high pressure (b) low temperature (c) low pressure (d) at any pressure
 - l. When a gas is compressed adiabatically, its temperature – (a) remains constant (b) increases (c) decreases (d) none of the above
 - m. The point that connects the saturated liquid line and saturated vapour line is called – (a) saturated point (b) critical point (c) triple point (d) superheated point

- n. Dryness fraction of dry saturated steam on heating – (a) increases (b) decreases (c) remain same (d) cannot predict
- o. Latent heat of vapourisation of steam – (a) increases with the increase of pressure (b) decreases with the increase of pressure (c) remains constant at all pressures (d) latent heat doesn't depend upon pressure
- p. Measure the dryness fraction of a very wet steam with the help of – (a) separating calorimeter (b) throttling calorimeter (c) combined separating and throttling calorimeter (d) bomb calorimeter
- q. The critical pressure of water is – (a) 7.38 MPa (b) 22.12 MPa (c) 100 MPa (d) 5.04 MPa
- r. The isentropic process on the Mollier chart is represented by – (a) horizontal line (b) vertical line (c) inclined line (d) curved line
- s. Equivalent evaporation is the amount of water evaporated in a boiler from and at – (a) 0°C (b) 100°C (c) saturation temperature at given pressure (d) room temperature
- t. Cochran boiler is a – (a) horizontal fire tube boiler (b) horizontal water tube boiler (c) vertical fire tube boiler (d) vertical water tube boiler
- u. A fusible plug is fitted in small boilers in order to – (a) control steam flow (b) extinguish fire if water level in boiler falls below alarming limit (c) remove ash (d) all the above <https://www.wbsctonline.com>
- v. The draught in boiler may be produced by – (a) mechanical fan (b) chimney (c) a steam jet (d) all of these
- w. _____ is NOT a boiler mounting. – (a) Blow off cock (b) Feed check valve (c) Economiser (d) Safety valve
- x. The water tubes in a Babcock and Wilcox boiler are – (a) horizontal (b) vertical (c) curved (d) inclined
- y. The maximum heat loss in a boiler occurs due to – (a) moisture in fuel (b) dry flue gases (c) steam formation (d) unburnt carbon

Group B

2.
 - a) A cylinder contains 0.45 m^3 of a gas at $1 \times 10^5\text{ N/m}^2$ and 80°C . The gas is compressed to a volume of 0.13 m^3 , the final pressure being $5 \times 10^5\text{ N/m}^2$. Determine: (i) The mass of gas; (ii) The value of index 'n' for compression; (iii) The increase in internal energy of the gas; (iv) The heat received or rejected by the gas during compression. Take $\gamma = 1.4$, $R = 294.2\text{ J/kg}^{\circ}\text{C}$.
 - b) Make a comparison of ideal gas with a real gas. 6+2
3.
 - a) Draw a Mollier diagram showing different lines.
 - b) The following observations were recorded in an experiment with a combined separating and throttling calorimeter:
 Pressure in the steam main: 15 bar
 Mass of water drained from the separator: 0.55 kg
 Mass of steam condensed after passing through the throttle valve: 4.20 kg
 Pressure and temperature after throttling: 1 bar, 120°C
 Using steam tables, evaluate the dryness fraction of the steam in the main, and state with reasons, whether the throttling calorimeter alone could have been used for this test. 2+6
4.
 - a) State Clausius statement of second law of thermodynamics. With the help of block diagrams, show the COP of refrigerator and COP heat pump.
 - b) A turbine operates under steady flow conditions, receiving steam at the following state: Pressure 1.2 MPa, temperature 188°C , enthalpy 2785 kJ/kg , velocity 33.3 m/s and elevation 3 m. The steam leaves the turbine at the following state: Pressure 20 kPa, enthalpy 2512 kJ/kg , velocity 100 m/s , and elevation 0 m. Heat is lost to the surroundings at the rate of 0.29 kJ/s . If the rate of steam flow through the turbine is 0.42 kg/s , calculate the power output of the turbine in kW. 3+5

5. a) Explain the steam generation process with the help of p-v and T-s diagram.
b) Define wet steam, dry saturated steam and superheated steam. 5+3
6. a) What do you mean by system? Classify systems and explain each system.
b) What do you mean by properties of a system? Define intensive properties and extensive properties with proper examples. 4+4
7. a) Prove that adiabatic process follows the law $PV^\gamma = \text{const.}$
b) A cylinder contains 0.02 m^3 of air at 1.66 MPa and 50°C . The air expands isothermally to 104 KPa . Determine: i) Work transfer, ii) Heat transfer, iii) Final volume. 4+4
8. a) List the different sources of heat losses in a boiler.
b) The following readings were taken during the test on a boiler for a period of 1 hour: steam generated 5000 kg ; coal burnt 700 kg ; CV of coal 31402 kJ/kg and quality of steam 0.92 . If the boiler pressure is 1.2 MPa and feed water temperature is 45°C , calculate the boiler equivalent evaporation and thermal efficiency. [using Steam Table] 2+6
9. a) Explain the construction and working principle of Lamont boiler with a neat sketch.
b) Distinguish between fire tube boiler and water tube boiler. 5+3
10. a) Derive the expression, $h = 353H \left[\frac{1}{T_a} - \left(\frac{m+1}{m} \right) \frac{1}{T_f} \right]$ of natural draught in mm of water column.
(All the symbols have their usual meanings)
b) List at least three mountings and three accessories used in boiler. 5+3

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